

Customer Retention Analysis Executive Summary

UCI Online Retail II Cohort Retention & RFM Segmentation

Background

The business behind this data [UCI Machine Learning Repository — Online Retail II](#) is a UK-based online retailer specializing in unique, all occasion gift and homeware items small novelty goods, party supplies, decorative pieces, and giftware, sold to both individual consumers and smaller wholesale buyers. The dataset spans December 2009 through December 2011, roughly two years of order level transaction history.

Gift and homeware purchases naturally recur birthdays, holidays, home refreshes which makes retention, not just one off sales, the real lever on long term revenue. This analysis exists to answer a question the business needs to act on: **who are our most valuable customers, are we keeping them, and where is that starting to break down?**

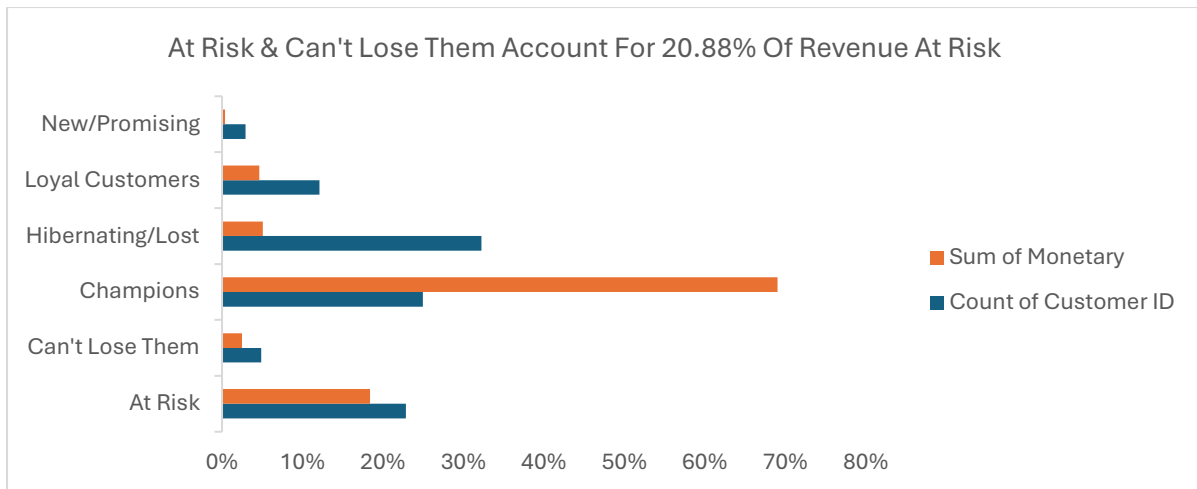
The Decision

Protect what's working and catch the revenue that's starting to slip.

A quarter of this business's customers its Champions rightly generate the majority of revenue, and that core is healthy. But a separate group of previously frequent buyers is showing clear signs of disengagement:

- **At Risk:** 22.86% of customers, 18.41% of revenue
- **Can't Lose Them:** 4.89% of customers, 2.47% of revenue
- **Combined:** 27.75% of customers $\approx$ **28%**, 20.88% of revenue $\approx$ **21%**

That's the group worth acting on now, before they join the third of the customer base (Hibernating/Lost, 32.24% of customers) that's already gone quiet and now contributes almost nothing (5.05% of revenue).



North Star Metrics

- **£17.4M** gross revenue over two years
- **72%** of customers bought more than once
- **69%** of revenue comes from Champions (25% of customers) a healthy, proportionate core
- **21%** of revenue sits with disengaging customers, At Risk + Can't Lose Them (28% of customers)
- **32%** of customers are already inactive (Hibernating/Lost), but only 5% of revenue large but low value
- **1 in 6** orders gets cancelled
- Retention has been **getting weaker, not stronger**, with every year the business has been open

Recommendations

Priority	Recommendation	Owner	Target	Metric to Watch
1	Launch a win-back campaign for At Risk + Can't Lose Them customers, starting with the 2009-12 cohort	Retention / CRM	Recover ≥15% of this group within one quarter	Reactivation rate, revenue recovered
2	Build an early warning trigger for frequent buyers whose activity starts	Retention / CRM	Reduce At Risk → Can't Lose	Segment migration rate,

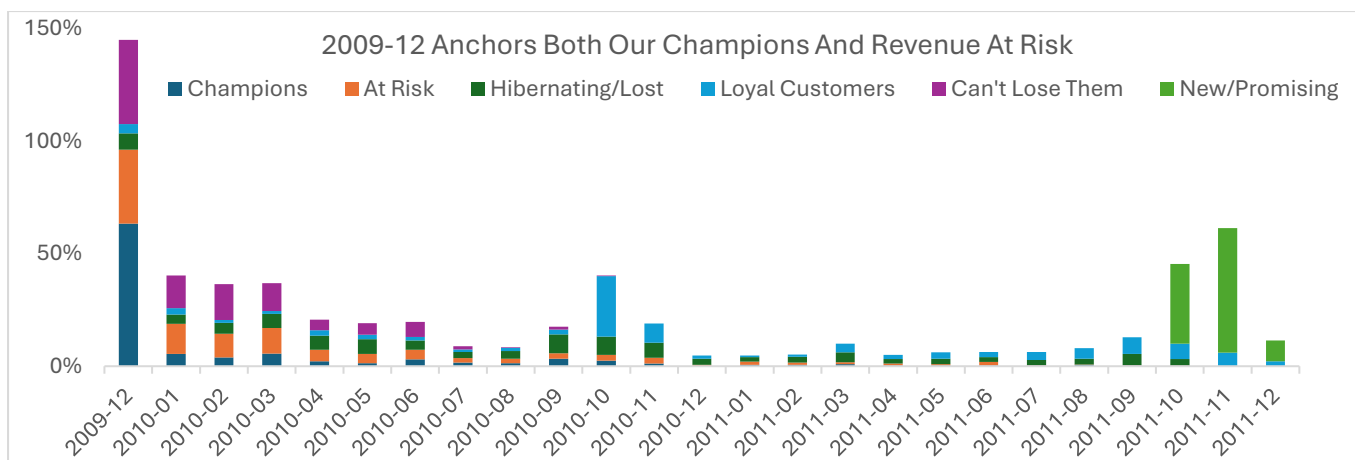
	slipping (before they become At Risk)		Them transition rate	month over month
3	Investigate what changed after 2010 new customer volume and retention both declined ~3x in parallel	Growth / Marketing	Identify root cause (channel, market, competition)	New cohort size, cohort retention ceiling
4	Strengthen onboarding and second purchase incentives for New/Promising customers	Marketing	Increase % making a second purchase within 60 days	Repeat purchase rate for new cohorts
5	Investigate the cancellation rate as an operational/product issue	Ops / Product	Reduce cancellation rate	Cancellation rate, cancellation reasons

The Story

Revenue concentration is largely healthy the core group is the right group.

Champions make up a quarter of customers and generate 69% of revenue. That's not a red flag; that's what a well functioning customer base looks like.

One early group of customers carries more than its share of both outcomes. The December 2009 cohort is disproportionately responsible for the business's best customers and disproportionately responsible for the customers now most at risk of being lost. That makes it the clearest starting point for the win back campaign in Recommendation 1.



The business is both acquiring fewer new customers and keeping them for less time. New customer volume dropped roughly threefold after the business's first year, and retention weakened across the same window. This dataset alone can't say why there's no marketing or acquisition channel data attached but the two trends moving together supports Recommendation 3's investigation.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
2009-12	991	36.91%	35.44%	45.32%	40.59%	37.85%	40.59%	36.91%	35.44%	39.12%	44.79%	52.47%	38.70%	29.55%	25.97%	31.86%	27.55%	32.49%	30.18%	27.66%	27.34%	33.75%	32.28%	43.43%	20.50%
2010-01	350	22.29%	33.14%	33.14%	26.57%	31.43%	27.14%	24.29%	29.14%	34.57%	33.43%	17.43%	22.29%	17.43%	19.71%	15.43%	24.29%	19.43%	19.14%	20.00%	25.71%	20.29%	24.29%	5.43%	
2010-02	356	23.31%	22.47%	29.21%	25.00%	19.94%	20.22%	27.81%	25.84%	28.37%	11.52%	12.64%	15.17%	17.98%	12.36%	20.51%	16.57%	16.57%	14.33%	23.03%	23.88%	16.85%	6.18%		
2010-03	421	18.29%	23.26%	23.75%	21.85%	20.19%	23.99%	29.93%	27.55%	11.16%	11.68%	14.25%	19.46%	16.63%	19.24%	16.86%	17.10%	14.96%	17.10%	20.19%	21.14%	7.36%			
2010-04	280	18.57%	18.21%	15.71%	17.86%	22.14%	27.50%	25.71%	10.71%	10.71%	7.50%	14.64%	13.93%	16.07%	15.36%	15.36%	14.29%	14.64%	17.50%	21.43%	5.71%				
2010-05	251	15.54%	16.73%	17.93%	18.33%	25.90%	20.72%	12.75%	5.98%	8.37%	11.55%	13.55%	14.74%	15.54%	9.96%	12.35%	13.94%	16.33%	15.54%	4.78%					
2010-06	259	18.15%	19.31%	20.46%	23.17%	28.96%	12.74%	9.27%	8.49%	11.58%	10.42%	13.90%	15.06%	12.36%	11.20%	12.36%	13.13%	20.85%	5.41%						
2010-07	182	15.93%	18.68%	28.57%	28.57%	14.29%	10.99%	14.84%	14.29%	11.54%	13.74%	14.84%	12.64%	13.74%	19.23%	17.58%	23.63%	8.24%							
2010-08	157	21.02%	30.57%	32.48%	17.83%	12.10%	10.19%	12.74%	14.01%	13.38%	13.38%	12.74%	15.92%	18.47%	17.83%	20.38%	7.01%								
2010-09	234	23.50%	24.36%	12.82%	9.40%	10.26%	14.10%	10.26%	12.82%	13.68%	11.97%	10.26%	22.22%	17.09%	16.67%	5.56%									
2010-10	370	25.68%	14.86%	12.70%	9.19%	8.38%	13.24%	14.32%	10.81%	9.46%	10.81%	12.70%	18.92%	23.24%	7.30%										
2010-11	319	17.87%	9.40%	9.72%	7.84%	8.78%	13.48%	10.03%	8.78%	9.09%	11.29%	14.73%	26.02%	6.27%											
2010-12	76	9.21%	5.26%	9.21%	11.84%	6.58%	5.26%	9.21%	5.26%	2.63%	9.21%	19.74%	2.63%												
2011-01	71	16.90%	21.13%	19.72%	21.13%	15.49%	15.49%	12.68%	11.27%	21.13%	25.35%	11.27%													
2011-02	123	16.26%	15.45%	18.70%	21.95%	15.45%	15.45%	14.63%	14.63%	17.89%	5.69%														
2011-03	178	17.98%	21.91%	19.66%	22.47%	15.17%	20.79%	20.22%	24.16%	9.55%															
2011-04	105	24.76%	20.00%	20.00%	18.10%	23.81%	18.10%	25.71%	11.43%																
2011-05	108	24.07%	25.00%	16.67%	22.22%	21.30%	25.93%	11.11%																	
2011-06	108	23.15%	21.30%	26.85%	20.57%	27.76%	8.33%																		
2011-07	101	20.79%	29.70%	27.72%	34.65%	15.84%																			
2011-08	106	26.42%	30.19%	26.42%	16.96%																				
2011-09	186	27.42%	37.63%	15.05%																					
2011-10	221	31.67%	15.84%																						
2011-11	192	14.06%																							
2011-12	28																								

Cancellations are common, not catastrophic yet. About 1 in 6 orders is cancelled. The revenue impact is modest (~4% of gross), but the frequency supports Recommendation 5.

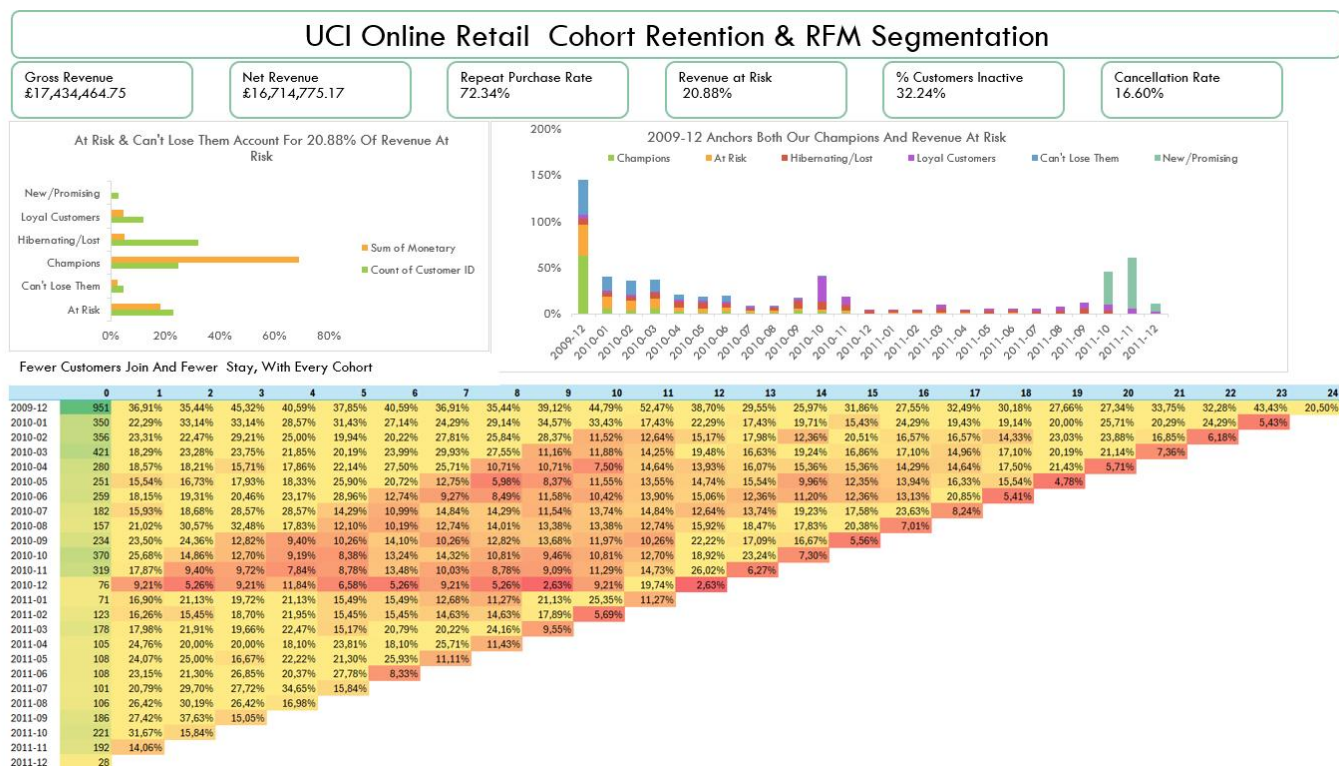
Financial Impact (*Estimate Directional, Not Validated*)

- **Revenue currently at risk:** roughly 20.88% of gross revenue, an estimated **£3.64M**, sits with customers already showing disengagement (At Risk + Can't Lose Them). This is money the business has *already earned* from these customers the risk is losing their *future* repeat business, not the £3.64M itself, which is historical, not projected.
- **Revenue already lost to cancellations:** £719,689.58 (4.1% of gross revenue) a direct, quantified figure, not an estimate, and independent of the segment story.
- **Revenue tied up in low-value inactive customers:** Hibernating/Lost customers (32% of the base) have generated an estimated **£880K-£900K** total (5.05% of £17.4M) useful context for *not* over-investing win-back budget here relative to the At Risk/Can't Lose Them group.

These figures are presented the way any early-stage finding should be as a starting estimate, not a guarantee. No specific recovery amount (e.g., "£X will be won back") is projected here, because that requires an assumed win back conversion rate, and no conversion rate exists yet for this business inventing one would make the number look more precise than it is. In practice, an estimate like this is only ever validated by running the actual campaign and measuring what comes back; the figures here are meant to justify *prioritizing* the At Risk/Can't Lose Them group over the Hibernating/Lost group (roughly triple the revenue concentration per customer), not to promise a specific

return. Once a pilot campaign runs, these estimates should be revisited and replaced with real observed figures.

Dashboard



Caveats & Limitations

- **The 2011-12 cohort was excluded** from all averages and current-retention figures. Its join count (28 customers) was dramatically lower than every other cohort (typically hundreds), and every rate in that period broke the surrounding pattern consistent with incomplete data capture near the dataset's end, not a genuine business signal.
- **"Active" in the retention table means "made a purchase at that point," not "purchased continuously with no gaps."** A customer can go quiet for several months and still count as active again once they return.
- **The Segment field is a fixed, lifetime classification** per customer, calculated once from their complete purchase history it does not change month to month or reflect what a customer looked like earlier in their journey.
- **A scoring bug was caught and corrected during this analysis:** Recency was initially sorted in the wrong direction, which silently inverted every customer's Recency score. It was caught because it produced a result that shouldn't have been possible brand-new customers appearing in the most disengaged segment

and was traced back and fixed before the segment figures in this report were finalized.

- **This dataset contains no marketing spend or acquisition channel data.**
Customer Acquisition Cost and forward-looking Lifetime Value could not be honestly calculated and are not included; the revenue at risk figures in this report are historical, observed totals, not predictions.
- **Financial impact figures are directional estimates**, not validated projections they are meant to guide prioritization, not promise a specific recovery amount.